

FARDIN SYED

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EDUCATION

PhD, University of Toronto

Sep 2022 - Present

Advisor: Duncan Dauvergne

Expected completion: 2027

Research area: Probability, KPZ universality

MSc, University of Toronto

Sep 2021 - Aug 2022

Mathematics

CGPA: 4.0/4.0

BSc, University of Toronto

Aug 2017 - May 2021

Mathematics and Physics

CGPA: 3.99/4.00

RESEARCH PAPERS

D. Dauvergne and **F. Syed**. *Bulk heights of the KPZ line ensemble*. arXiv:2602.06876, 2026.

A. M. Stock, **F. Syed**, M. H. van Kerkwijk, R. Lin, F. Kirsten, and U.-L. Pen. *Scintillation properties of PSR B1133+16 measured with very long baseline interferometry*. *The Astrophysical Journal* 992(2), 192, 2025.

F. Syed, R. D. Sipio, and P. Sinervo. *Bidirectional long short-term memory (BLSTM) neural networks for reconstruction of top-quark pair decay kinematics*. 2019.

INVITED TALKS

Probability Seminar, University of Illinois Urbana-Champaign, Winter 2027.

Probability Seminar, Indiana University Bloomington, Fall 2026.

How SPDEs May Be Used to Construct Euclidean QFTs, Geometric Structures Laboratory, Fields Institute, Oct 2025.

The KPZ Line Ensemble Converges to Positive Infinity, Probability Seminar, Fields Institute, Apr 2025.

The Longest Increasing Subsequence of a Random Permutation, University of Toronto, Department of Mathematics, Graduate Student Seminar, Dec 2023.

WORKSHOPS AND SUMMER SCHOOLS

AIM Workshop on Gibbsian Line Ensembles, California Institute of Technology

Aug 2026

School on Stochastic Interacting Particle Systems and Random Matrices, Rényi Institute

Jun 2025

Summer School in Stochastic Quantization, Simons Laufer Mathematical Sciences Institute

Jul 2024

TEACHING AND MENTORING

Course Instructor - University of Toronto

- Mathematical Probability - MAT377 *Winter 2026*

Reading Course Instructor - University of Toronto

- Random Matrix Theory *Summer 2026*

Teaching Assistant - Simons Laufer Mathematical Sciences Institute

- Random Growth Models, Phase Separation and Hamilton-Jacobi PDE *Jun 2026*

Teaching Assistant - University of Toronto

- Real Analysis - MAT357 *Winter 2025*
- Mathematical Probability Theory - MAT377 *Fall 2024*
- Probability I and II - MAT1600 & MAT1601 *2022–2023*
- Multivariable Calculus with Proofs - MAT237 *2021 - 2024*
- Linear Algebra II - MAT224 *Fall 2020*

ACADEMIC SERVICE

Referee, *The Annals of Probability* *2026*

Organizer, *Bird's Eye View Conference* *Mar 2025, Jan 2026*

Organizer, *Math & Computer Science Symposium* *Mar 2025*

Organizer, *Mathematics Graduate Student Seminar* *Sep 2024 - Apr 2025*

SCHOLARSHIPS AND AWARDS

Queen Elizabeth II Graduate Scholarship in Science and Technology *Sep 2026 - Aug 2027*

Ontario Graduate Scholarship *Sep 2024 - Aug 2026*

Ida Bulat Memorial Graduate Fellowship *May 2023*

Beatrice Evelyn Rodgers Scholarship *Mar 2022*

The Coxeter Scholarship in Mathematics *Nov 2020*

Boris Stoecheff Memorial Scholarship in Physics *Nov 2020*

Undergraduate Student Research Award, NSERC Canada *May 2019*